ACHRAF RAHIBE

UAV & AVIONICS SYSTEMS ENGINEER



Rabat, Morocco | +212 620 323 768
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linkedin.com/in/achraf-rahib1 | Open to relocation

Integrates flight controls, embedded electronics and power systems across aerial, ground and maritime platforms. Leads electronics coordination and delivers configuration, troubleshooting and test support for engineering teams and clients.

TECHNICAL EXPERTISE

Flight control
ArduPilot, PX4, INAV; MAVLink; Mission Planner, QGroundControl; SITL, PID, logs and FFT/filtering.

Embedded systems
C/C++, Python, VHDL; STM32, ESP32, Raspberry Pi, Jetson; Linux.

Avionics interfaces
Pixhawk, Cube, CUAV; GNSS/RTK, DroneCAN, UART, SPI, I2C, Ethernet and telemetry.

Electronics and power
Li-ion/BMS, ESC/VESC, wiring, pre-charge/contactors; Altium/KiCad familiarity.

Vision and RF
YOLOv8, ByteTrack, RTSP/GStreamer, SIYI; RTL-SDR and AIS.

Industrial and CAD
Siemens PLCs, TIA Portal, WinCC, VFDs; AutoCAD Electrical, Onshape, 3D printing.

EDUCATION

State Engineer Diploma
Electrical Engineering and Telecommunications; Industrial Control.
FSTM, Hassan II University.
2021-2025 | Bac+5

DUT in Electrical Engineering
Electronics and Industrial Computing.
EST Fès, Université Sidi Mohamed Ben Abdellah.
2019-2021

Technical Baccalaureate
Electrical Sciences and Technology.
Lycée Technique Anoual, Casablanca.
2018-2019

LEADERSHIP

ROBOTHINGS | Founder and President, 2023-2025.
ENACTUS FSTM | Marketing Lead, 2022-2023.

LANGUAGES

Arabic: Native
English & French: Fluent
German: Basic

PROFESSIONAL EXPERIENCE

FARASHA SYSTEMS | Rabat, Morocco

Electronics Team Manager | Mar 2026 - Present

Lead avionics integration, component selection and power/payload interfaces. Coordinate electronics, embedded, RF and mechanical teams; review solutions, plan validation and resolve integration issues.

UAV & Avionics Systems Engineer | Aug 2025 - Mar 2026

Integrated multirotor, fixed-wing, VTOL and USV systems. Configured navigation, sensors, telemetry, missions and failsafes; used simulation, bench checks and logs to prepare ground/flight trials.

UAV & Robotics Engineer | 2025 - Present

Independent engineering | Upwork, Fiverr and direct clients | Remote

Deliver autopilot configuration, stability tuning and log analysis for client platforms. Translate faults into corrective actions, diagnostic reports and practical validation plans.

CAPGEMINI ENGINEERING | Casablanca

Technical Documentation Technician | Jul - Sep 2023

Produced Stellantis after-sales diagnostic and service documentation from automotive engineering data.

ENGINEERING PROJECTS

Aircraft and avionics integration
Integrated fixed-wing, VTOL and multirotor systems; worked on flight computers, payload interfaces, propulsion sizing, high-current wiring and 14S / Li-ion/BMS architecture concepts.

Flight control and diagnostics
Configured autopilots, sensors, motor/servo mapping and failsafes. Analysed PID response, vibration and helicopter logs; supported successful VTOL trials and reported static run-up findings.

Marine and ground robotics
Integrated 4 m twin-waterjet USV electronics, ESC/BMS and pre-charge interfaces; configured CUAV X7 / dual-VESC rover control and explored platform-coordination concepts.

Embedded perception and communications
Integrated Jetson/Raspberry Pi, tracking and SIYI video interfaces; worked on GNSS/RTK, telemetry and AIS reception, logging and replay.

UAS engineering delivery
Planned drone component compatibility, BOMs, workshop tools and spares; translated configuration and log-analysis findings into technical reports, corrective actions and validation plans.